

ASEN 5264 Decision Making under Uncertainty

Exam 3: POMDPs and Games

April 8, 2026

Instructions: Clearly indicate your final answers and **briefly justify all answers with text, mathematical expressions, and/or calculations**. If you do not understand how to do a problem, skip it and move on so that you have time to attempt all problems. You may consult any static source, but you may NOT communicate with any person except the instructor or TA, and you may not use LLMs such as ChatGPT.

Important: Please include the following statement along with your signature (typed or handwritten) on the exam: **“I affirm that I completed this exam without the use of any artificial intelligence tool such as ChatGPT or Github Copilot.”**

Note: Question 1 and 2 have been removed from this exam because they were added to the homework.

Question 1. (10 points) Suppose that Alice and Bob are playing the 5-card version of “Goofspiel” or “Game of Pure Strategy” discrete Markov game played in class. Alice has computed a Nash equilibrium for the game and she shares her Nash equilibrium policy, π_A^1 , with Bob.

- a) If Bob knows the (I, S, A, T, R, γ) model of the game and Alice’s policy π_A^1 , describe in two or three sentences and equations how Bob can compute a best response policy with a computer.

Solution: Bob can form an MDP using the given model parameters $M' = (S, A^{Bob}, T', R', \gamma)$ with

$$T'(s'|s, a^{Bob}) = \sum_{a^{Alice}} T(s'|s, a^{Bob}, a^{Alice}) \pi_A^1(a^{Alice}|s)$$

and

$$R'(s, a^{Bob}) = \sum_{a^{Alice}} R^{Bob}(s, a^{Bob}, a^{Alice}) \pi_A^1(a^{Alice}|s)$$

and solve the MDP for the best response policy using any of the MDP solvers we have covered in this course.

- b) If Bob then plays the best response policy computed in the manner described above against π_A^1 , is the resulting joint policy guaranteed to be a Nash equilibrium? Briefly justify your answer.

Solution: No. Though Bob’s policy is a best response to Alice’s, Alice’s policy may not be a best response to Bob’s. For example, even though the Nash equilibrium (shown in class) consists of mixed policies, Bob may calculate a deterministic best response. Then Alice may have a best response to that policy that is not a Nash equilibrium policy.